Claude Code change request v5: catalogue, names, specialisations, readiness check

Paste everything below into the Claude Code session that is building Steady Strength. It extends the existing build (importer, coach_decisions.json, library_catalogue.py, IMPORT_REPORT.md). Do not rewrite what already works. Attached files: `catalogue_draft.json`, `volume_audit.csv`, `module_compatibility_matrix.csv`, `AUDIT_REPORT.md`, `ADDENDUM_missing_deliverables.md`.

Ground rules

1. **Audit before coding.** First read the attached files and the existing repo. Reply with a short plan and the list of existing files you will change. Then build.
2. **Preserve originals.** Never overwrite source spreadsheets or earlier imports. New content goes in new versioned files.
3. **Everything in `catalogue_draft.json` is DRAFT_UNVERIFIED.** Import it with `status = draft` and `reviewed = false`. Show a visible "Draft, not reviewed" flag in an internal/debug build, and block it from the public catalogue behind a `contentApprovedForRelease` flag that defaults to false. Do not present it as final production data.
4. Keep existing behaviour for accounts, Face ID, privacy, calories, check-in, HealthKit. This request only touches training content and the engine.

1. Catalogue (replace the earlier program list)

Ship **five core programs** (12 weeks) from `programs[]`: Strength Foundations (3d), Build & Grow (4d), **Lower Emphasis** (4d, master name "Lower Body Focus"), **Muscle Builder** (5d, master name "Muscle Builder Pro"), Strong for Life (2-3d, third day optional).

- Show proposed names. Keep `master_name` as an alias for search and for mapping old imports.
- A names-validator (unit test): every program, template, module and exercise display name is unique.

- The earlier Beginner Options A-D are not imported. The Appendix A perimenopause pathway (Strong Foundations, Stronger Every Week, Strength & Shape, Strength Plus, The Strength Routine; working family name "Strong Through Midlife") is a **separate pathway family**, authoritative in its own .docx. Do NOT import it from the master PDF (no sets/reps there). Create the schema and an empty `PathwayFamily`, and wait for the .docx; never invent its prescriptions. Keep the 28-exercise cue library and the Learn-phase rule (weeks 1-4, 2 sets at RPE 6-7) as the beginner on-ramp.
- Phase labels Find Your Strength (1-4), Build Your Strength (5-8), Feel Stronger (9-12) belong to the Strong Through Midlife pathway only.
- **Coach Series** (RW Fitness) stays a separate section built from the existing importer. Unchanged except renames in the draft file. Items in `coach_decisions.json` still need Riki's answers.

2. Specialisation modules

Ten modules from `specialisations[]` (6 lower, 4 upper; upper display names Arm Builder, Back Builder, Chest Builder, Shoulder Builder), 6 weeks, 2 exposures per week. Rules (all configurable in a `RuleSet` table, not hard-coded):

1. One active module by default. A second needs explicit confirmation and shows combined weekly sets by muscle with an overlap warning.
2. Activation **overlays** the base program for 6 weeks: the module's A/B sessions are merged into two receiving sessions; the base week pointer pauses. (Lower modules: the module's two sessions replace the base program's lower sessions. Needs coach confirmation.)
3. Merge algorithm: (a) remove base **accessory** slots whose direct muscles are inside the module's target muscles; (b) add module slots; (c) if estimated duration exceeds the user's limit (default 60 min incl. warm-up/cool-down) trim the lowest-priority remaining accessory until it fits; (d) show the user a "What changed" list (added, removed, trimmed). Never silently add sets.
4. Never offer upper modules by default on Strength Foundations or Strong for Life. Beginners stay on two lower sessions.
5. Activation is idempotent: activating twice never creates duplicate sets (acceptance test 3).
6. Deactivation restores the base session exactly (acceptance test 8).
7. Volume tracking: store **direct** and **indirect** sets separately. Compounds credit the primary mover 1.0 and secondaries 0.5 (configurable). A weekly volume screen labels figures "approximate". "Excessive volume" triggers a **review prompt**, not a

block (test 9). Thresholds are placeholders: review prompt when direct hard sets for one muscle exceed 20 per week, to be confirmed by the coach.

8. Week modifiers come from `week_modifiers` in the draft. Lower and upper modules currently disagree about a planned week 4; implement ONE policy: a planned lower-fatigue week is a *suggestion* the user can accept or decline, informed by readiness and performance. Never silent.

3. Exercise data

- Import 90 canonical exercises from `exercises[]`. Keep `source_names[]` as aliases.
- Each exercise: `id, name, direct_muscles, indirect_muscles, movement_pattern, equipment, unilateral, unit (reps/seconds), per (leg/side), alternatives, review_flags, video`.
- Items flagged AMBIGUOUS get the default picks listed in `AUDIT_REPORT.md` section 6, marked `needs_coach_confirmation`.
- Substitutions: use `approved_starting_substitutions`. A swap creates a **new exercise-specific baseline**; history stays on the original exercise ID (test 5). Show "not necessarily equivalent" text. If the user reports pain, do not claim any alternative is safe; show the "get individual assessment" prompt.
- `load_increment_kg` is null in the draft. Use a settings default (2.5 kg upper, 5 kg lower) until the coach sets per-exercise values. Show increments as editable.
- Review flags: Nordic curl (assisted), trap-bar deadlift, back squat 4x4-6 and front squat get `contraindication_review_flag = true` and an easier-regression suggestion.

4. Videos

No video is approved. For every exercise set `video.status = none_approved` and use the `demoSearchQuery` fallback (opens search in the browser; no embedding, no downloading, no rehosting). Add a `ExerciseMedia` entity with `source, licence, url, approved, verifiedBy, verifiedAt`. Only `approved = true` items play in-app. User-recorded clips via PHPicker are private to the user.

5. Data model (17 entities, SwiftData)

Program, ProgramVersion, Phase, WeekPrescription, WorkoutTemplate, ScheduledSession, Exercise, ExerciseAlternative, ExerciseMedia, Specialisation, CompatibilityRule, WeeklyVolumeTarget, UserEnrolment, ExerciseLog, ReadinessCheck,

ProgressionRecommendation, AuditTrail.

- **Published prescriptions are immutable.** A content update creates a new `ProgramVersion`; a user's `UserEnrolment` stays pinned to its version unless they opt to move (test 10).
- **Logs are mutable only by the user**, with edits recorded in `AuditTrail`.
- Slots carry `sets, rep_min, rep_max, unit, per, rest_seconds, target_RIR (from week modifier), load_increment_kg`.
- Add lightweight migration from the current schema. Do not delete existing user data.

6. Readiness check (new, optional)

Before a session, an optional 3-question check: sleep, energy, anything that bothers you today (free choice, no diagnosis). Offers: do as planned, lower load, fewer sets (about 30-50%), comfortable substitute, or rest. It never diagnoses, never blocks, and the answers are private, off by default for cloud sync, deletable, and covered by the existing consent screen. Wording avoids medical claims. Pain or dizziness shows: "Stop and speak to a doctor or qualified professional."

7. Progression and scheduling

- Double progression as in the master: if all sets hit rep_max at or above the target RIR with acceptable technique, suggest the smallest practical increase; otherwise hold. **Suggestions need user confirmation** and can be declined.
- Substantial drop across two exposures: suggest hold or reduce and review sleep, food, technique.
- Missed session: keep sequence, shift the date, offer reschedule. Never double up hard sessions. A week does not advance because 7 days passed.
- Returning after missed weeks (existing rule): 2 weeks missed, go back one week at about 90%; 3+ weeks or illness, week 1 at about 80%. Mark as coach-proposed, not clinical.
- Optional workouts are labelled optional and never count as missed.

8. Acceptance tests (write them as unit/integration tests)

The master's ten, rewritten where the original cannot pass:

1. **Rewritten:** all 19 Appendix B templates and 10 modules import with every slot resolving to an exercise ID and having sets, rep_min, rep_max (the "240

assignments" test is dropped until Appendix A has prescriptions).

2. Every 6-week module has both A and B sessions and week modifiers 1-6.
3. Activating a module twice creates no duplicate working sets.
4. Plank/farmer carry show seconds; lunges show "per leg"; dead bug/Pallof show "per side".
5. A substitution does not overwrite historic loads.
6. Missed sessions resume in order.
7. **Rewritten:** base program week modifiers (weeks 4 and 8 optional lower-fatigue) apply only if accepted by the user.
8. Specialisations can be turned off and the base session is restored exactly.
9. Excess weekly volume triggers a review prompt.
10. Publishing a new ProgramVersion preserves all historical logs.
    Plus: names validator (section 1), merged session never exceeds the user's time limit, no draft content appears in a release build.

9. Do not do

- Do not invent sets, reps, increments or videos. Missing values stay null and are listed in the import report.
- Do not claim medical, clinical or evidence-based outcomes in app text. No "2026 ACSM position stand" or other citation until verified.
- Do not copy Runna, Strava or commercial program content.

10. Report back

Update `IMPORT_REPORT.md` with: counts imported (expect 5 programs, 19 templates, 10 modules, 90 exercises), items needing coach confirmation, null fields, and any test you could not make pass.